

UNIT- V

HOUSING IN DISASTER PRONE AREAS

Introduction:

Housing in disaster-prone areas presents unique challenges due to the heightened risk of natural disasters such as earthquakes, hurricanes, floods, wildfires, and other catastrophic events. The design, construction, and planning of housing in these areas require special considerations to enhance resilience, minimize vulnerability, and ensure the safety of residents. This introduction explores the key aspects of housing in disaster-prone areas.

Earthquake in disaster prone areas:

Earthquakes are a significant natural hazard that can cause widespread destruction, particularly in areas prone to seismic activity. When planning for and responding to earthquakes in disaster-prone areas, various considerations must be taken into account to enhance preparedness, minimize vulnerability, and facilitate effective response and recovery efforts. Here are key aspects related to earthquakes in disaster-prone areas:

Damages to houses:

Houses can suffer various types of damages due to a range of factors, including natural disasters, accidents, and long-term wear and tear. The extent and nature of the damage can vary widely depending on the specific cause. Here are some common causes of damage to houses:

1. Natural Disasters:

- **Earthquakes:** Structural damage, foundation issues, and collapsing walls.
- **Floods:** Water damage to foundations, walls, and possessions.
- **Hurricanes/Cyclones:** Wind damage, roof and siding destruction, and flooding.
- **Wildfires:** Destruction of homes through burning or exposure to extreme heat.

2. Water Damage:

- **Leaking Roofs:** Water infiltration causing damage to ceilings, walls, and insulation.
- **Burst Pipes:** Flooding, water damage to walls and flooring.
- **Flooding:** Overflowing rivers, storm surges, or heavy rainfall leading to extensive water damage.

3. Fire:

- **Structural Damage:** Burning of building materials, collapse of walls or roof.
- **Smoke Damage:** Residue and odor permeating walls, furniture, and belongings.

4. **Structural Issues:**

- **Foundation Problems:** Settlement, cracks, or shifting leading to structural instability.
- **Roof Damage:** Leaks, missing shingles, or collapse.

5. **Mold and Mildew:**

- **Water Intrusion:** Persistent dampness leading to mold and mildew growth.
- **Poor Ventilation:** Lack of proper airflow allowing mold to thrive.

6. **Pests and Infestations:**

- **Termites:** Structural damage through the consumption of wood.
- **Rodents and Insects:** Nesting, gnawing, and damage to walls, insulation, and wiring.

7. **Accidents:**

- **Impact Damage:** Collisions from vehicles, fallen trees, or other objects.
- **Structural Failure:** Poor construction leading to collapses or failures.

8. **Wear and Tear:**

- **Aging:** Deterioration of materials over time.
- **Poor Maintenance:** Lack of regular upkeep leading to various issues.

9. **Expansive Soils:**

- **Soil Movement:** Swelling or shrinkage of soil causing foundation issues.

10. **Human Factors:**

- **Vandalism:** Intentional damage to property.
- **Negligence:** Improper use or maintenance causing preventable damage.

TRADITIONAL PRONE AREAS:

It seems there might be a slight typo in your question. Did you mean "traditionally prone areas"? If so, I'll provide information based on that assumption. If you meant something else, please provide clarification.

Assuming you're referring to "traditionally prone areas," this could relate to regions or communities that have historically been susceptible to certain challenges, patterns, or conditions. Here are a few contexts in which the term might be used:

1. Natural Disasters:

- Areas that have a history of being prone to specific natural disasters, such as earthquakes, hurricanes, floods, or wildfires. These regions may experience these events more frequently due to their geographical location or climatic conditions.

2. Economic Vulnerability:

- Certain regions may be traditionally prone to economic challenges, such as poverty or unemployment. Factors like limited job opportunities, lack of diverse industries, or historical economic downturns can contribute to a persistent vulnerability.

3. Health Challenges:

- Regions with a historical prevalence of specific health issues, whether due to environmental factors, limited access to healthcare, or other circumstances. These areas may face challenges related to the spread of diseases or inadequate healthcare infrastructure.

4. Social Vulnerabilities:

- Communities that have been historically prone to social challenges, such as discrimination, inequality, or lack of access to education. These issues can persist over time and affect the overall well-being of the population.

5. Environmental Degradation:

- Areas prone to environmental degradation due to factors like deforestation, overgrazing, or unsustainable agricultural practices. Such regions may experience ongoing challenges related to soil erosion, loss of biodiversity, and other environmental issues.

6. Conflict Zones:

- Regions with a history of conflict or political instability may be traditionally prone to social and economic disruptions. These areas may struggle with issues such as displacement, human rights violations, and challenges in rebuilding after conflicts.

7. Climate Change Impact:

- Areas that are traditionally prone to the impacts of climate change, such as rising sea levels, changing precipitation patterns, or increased temperatures. These regions may face challenges related to adapting to new climatic conditions.

TYPE OF DAMAGES AND RAILWAYS OF NON-ENGINEERED BUILDINGS:

Non-engineered buildings, also known as vernacular or traditional buildings, often lack the structural design and construction standards employed in engineered structures. As a result, they may be more vulnerable to various types of damages, particularly during natural disasters or adverse events. Here are common types of damages that non-engineered buildings, including those near railways, may experience:

1. Collapse:

- Non-engineered buildings may be more susceptible to total or partial collapse, especially during seismic events or extreme weather conditions. This can pose a significant risk to the occupants and surrounding infrastructure.

2. Foundation Failure:

- Inadequate foundation design or construction may lead to settlement, shifting, or failure, affecting the overall stability of the building. This is a common issue in non-engineered structures.

3. Roof Damage:

- Non-engineered buildings may have roofs that are more prone to damage, including leaks, missing or damaged roofing materials, and inadequate resistance to wind loads. This vulnerability can lead to water infiltration and structural problems.

4. Wall Failures:

- Walls in non-engineered buildings may lack proper reinforcement and be more susceptible to cracking, leaning, or collapsing. This can occur during earthquakes, storms, or other events that subject the building to lateral forces.

5. Poor Material Quality:

- Non-engineered buildings often use locally available materials without rigorous quality control. This can result in substandard construction, making the structure more susceptible to decay, rot, and overall material deterioration.

6. Vulnerability to Fire:

- Non-engineered buildings may lack fire-resistant materials and design features, increasing their vulnerability to fire incidents. Proximity to railways can elevate the risk if there are train-related accidents or incidents involving hazardous materials.

7. Inadequate Seismic Resistance:

- In seismic-prone regions, non-engineered buildings may lack proper seismic design features, making them more susceptible to damage or collapse during earthquakes.

8. Lack of Compliance with Building Codes:

- Non-engineered buildings may not adhere to established building codes and standards, resulting in structures that are not well-equipped to withstand various environmental stressors.

9. Accessibility Challenges:

- Non-engineered buildings may lack proper accessibility features, making them less suitable for individuals with disabilities. This can be a critical concern for emergency evacuation during railway incidents.

10. Environmental Impact:

- Non-engineered buildings may contribute to environmental degradation if constructed without proper consideration for sustainable practices. This can include deforestation, soil erosion, and other negative environmental effects.

REPAIR AND RESTORE ACTION OF EARTHQUAKE-DAMAGED NON-ENGINEERED BUILDINGS:

1. Structural Assessment:

- Conduct a thorough structural assessment of the damaged buildings to understand the extent of the damage and identify critical issues. Engage structural engineers or experts in seismic retrofitting to evaluate the condition of the building.

2. Priority Repairs:

- Prioritize repairs based on the severity of damage and the potential risk to occupants. Immediate safety concerns, such as unstable structures or hazardous conditions, should be addressed first.

3. Seismic Retrofitting:

- Consider implementing seismic retrofitting measures to enhance the earthquake resistance of the buildings. Retrofitting may include adding shear walls, bracing systems, or other structural elements to improve the building's ability to withstand seismic forces.

4. Foundation Strengthening:

- Strengthen the building's foundation to improve its resistance to settlement and shifting during earthquakes. This may involve underpinning or adding additional supports to enhance stability.

5. Roof and Wall Repairs:

- Address roof and wall damage, repairing or replacing components as necessary. Strengthen walls and roofs to resist lateral forces and prevent collapse during future seismic events.

6. Use of Local Materials:

- Utilize locally available and culturally appropriate materials for repairs. Ensure that the materials are of high quality and meet safety standards.

7. Community Involvement:

- Involve the community in the repair and restoration process. Local knowledge and expertise can be valuable, and community engagement fosters a sense of ownership and resilience.

8. Education and Training:

- Provide training and education to local communities on earthquake-resistant construction techniques. This empowers residents to participate in future construction projects and adopt safer building practices.

RECOMMENDATIONS FOR FUTURE CONSTRUCTIONS:

1. Building Codes and Standards:

- Develop or enhance building codes and standards that specifically address earthquake-resistant construction for non-engineered buildings. Implement and enforce these codes to ensure safer structures in the future.

2. Engineered Support:

- Integrate engineering principles into the construction of non-engineered buildings. This may involve working with local builders to incorporate simple yet effective engineering techniques for seismic resistance.

3. Seismic Design Guidelines:

- Provide clear and accessible seismic design guidelines for builders and architects in earthquake-prone areas. These guidelines should be tailored to the specific challenges and characteristics of non-engineered construction.

4. Community Planning:

- Integrate earthquake-resistant design principles into community planning. Consider the layout of buildings, open spaces, and infrastructure to enhance overall seismic resilience.

5. Local Capacity Building:

- Build local capacity by training builders, contractors, and architects in earthquake-resistant construction techniques. This helps ensure that the knowledge is embedded within the community for future projects.

6. Incentives for Compliance:

- Offer incentives or support for builders and homeowners who adhere to seismic-resistant construction practices. This can encourage compliance with safety standards.

7. Public Awareness Campaigns:

- Launch public awareness campaigns to educate residents about the importance of earthquake-resistant construction and safety measures. This includes information on building practices, emergency preparedness, and evacuation plans.

8. Research and Innovation:

- Invest in research and innovation to develop cost-effective and culturally sensitive earthquake-resistant building solutions for non-engineered constructions.

REQUIREMENTS OF STRUCTURAL SAFETY OF THIN PRECAST ROOFING UNITS AGAINST EARTHQUAKE FORCE

Thin precast roofing units, such as concrete slabs or panels, are vulnerable to seismic forces, and ensuring their structural safety during an earthquake is crucial. The structural safety of these elements can be addressed by considering several key requirements:

Site-Specific Seismic Hazard Assessment:

- Conduct a site-specific seismic hazard assessment to understand the seismic forces that the structure may encounter. This involves evaluating the seismicity of the region, potential ground accelerations, and other geological factors that influence earthquake forces.

Structural Analysis and Design:

- Engage structural engineers to perform a thorough analysis and design of the thin precast roofing units. The design should consider the anticipated seismic forces, the flexibility and stiffness of the units, and compatibility with the rest of the structure.

Reinforcement and Material Properties:

- Ensure that the materials used in the precast roofing units, such as concrete, are of appropriate quality and that the units are adequately reinforced to withstand seismic forces. The reinforcement layout should be designed to provide ductility and energy dissipation during an earthquake.

Quality Control and Inspection:

- Implement strict quality control measures during the manufacturing and installation of thin precast roofing units. Regular inspections should be conducted to verify that the units meet design specifications.

Emergency Access and Evacuation:

- Consider the impact of seismic forces on emergency access and evacuation routes associated with the roofing system. Ensure that any potential hazards, such as falling debris, are minimized.

Continued Monitoring and Maintenance:

- Implement a system for continued monitoring and maintenance of the roofing units to address any issues that may arise over time. Regular inspections can help identify and rectify potential vulnerabilities.

STATUS OF R&D IN EARTHQUAKE STRENGTHENING MEASURES - FLOODS, CYCLONE, FUTURE SAFETY**EARTHQUAKE STRENGTHENING MEASURES:****1. Innovative Retrofitting Techniques:**

- Researchers are exploring and developing innovative retrofitting techniques to enhance the seismic resilience of existing structures. This includes the use of advanced materials, base isolators, and energy dissipation devices.

2. Performance-Based Design:

- The shift towards performance-based seismic design is gaining momentum. Researchers are working on developing methodologies that focus on achieving specific performance objectives rather than simply meeting code-prescribed requirements.

3. Smart Infrastructure:

- Integration of smart technologies, such as real-time structural health monitoring, to assess the performance of structures during and after earthquakes. This helps in understanding the dynamic behavior of structures and aids in decision-making for emergency response and recovery.

4. Community Resilience:

- R&D efforts are also directed towards community-level resilience, including the development of early warning systems, community preparedness strategies, and the integration of social science aspects into earthquake risk reduction.

FLOODS:

1. Flood Risk Mapping and Modeling:

- Advanced mapping and modeling techniques are being developed to better understand and predict flood risks. This includes the use of satellite imagery, remote sensing, and Geographic Information System (GIS) technologies.

2. Nature-Based Solutions:

- Research is focusing on nature-based solutions for flood management, such as the restoration of wetlands, creation of buffer zones, and the use of green infrastructure to mitigate flood impacts.

3. Resilient Infrastructure:

- Developing flood-resilient infrastructure designs and materials to minimize damage and enhance the recovery of communities after flood events.

4. Climate Change Adaptation:

- Considering the impact of climate change on flood patterns and developing adaptive strategies to address changing precipitation and sea-level rise.

CYCLONES:

1. Improved Cyclone Forecasting:

- Advances in meteorological research are improving the accuracy of cyclone forecasting, allowing for better preparedness and evacuation planning.

2. Resilient Building Designs:

- Research on cyclone-resistant building designs, including the use of wind-resistant materials, improved roof structures, and building codes that account for cyclonic winds.

3. Community Education:

- R&D efforts include developing effective community education programs to ensure that residents in cyclone-prone areas understand the risks and are prepared to take appropriate actions.

FUTURE SAFETY:

1. Climate Resilience:

- Considering the broader aspects of climate resilience in urban and rural planning, including measures to address multi-hazard scenarios and the compounding effects of different disasters.

2. Technological Innovations:

- Ongoing R&D in emerging technologies, such as artificial intelligence, machine learning, and advanced sensing, to improve early warning systems, risk assessments, and disaster response.

3. Policy and Governance:

- Research on effective policies and governance structures to promote disaster risk reduction, enhance community resilience, and ensure sustainable development in the face of changing climate patterns.

4. Interdisciplinary Approaches:

- Increased emphasis on interdisciplinary research that integrates engineering, social sciences, and policy studies to develop holistic and effective solutions for future safety.